

Itching

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Itching can be very uncomfortable. It is one of the most common reasons people see doctors who specialize in skin disorders (dermatologists).

Scratching temporarily relieves itching but can damage the skin, sometimes resulting in more itching (the itch-scratch cycle) or infection (called a secondary infection). Over time, the skin can become thick and scaly (called lichenification).

Causes of Itching

Itching can result from:

- Skin disorders
- Disorders of other organs (systemic disorders)
- Disorders of the nervous system
- Mental health conditions
- Drugs, medications, and chemicals

In many skin and systemic allergic disorders, itching is caused by histamine. Histamine is a chemical in the body that is stored in mast cells and is involved in allergic reactions. **Mast cells** are part of the **immune system**. When mast cells are stimulated by various allergens (substances that cause an allergic reaction), they release histamine. When histamine is released into the blood, it can cause many symptoms, including itching and inflammation. There are a number of other chemicals in the body that play an important role in itching, such as proteases and **cytokines**.

Skin disorders

The **most common causes of itching** are skin disorders:

- Dry skin (see **xeroderma** and **ichthyosis**)

- [Atopic dermatitis](#) (sometimes called eczema)
- [Allergic contact dermatitis](#) (an allergic rash resulting from direct contact with a particular substance)
- [Hives](#) (can be caused by a variety of skin or systemic conditions or medications and chemicals)
- [Fungal skin infections](#)
- Parasitic skin infections (such as [scabies](#))
- [Insect bites](#)

Systemic (bodywide) disorders

Several systemic disorders can cause itch without any visible changes on the skin (such as a rash). Systemic disorders are less common causes of itch than skin disorders.

Some of the **more common systemic causes of itching** are:

- [Chronic kidney disease](#)
- [Diabetes](#)
- Disorders of the liver or gallbladder, particularly when they cause [jaundice](#) (yellowing of the skin)
- [Hyperthyroidism](#) (an overactive thyroid gland)
- [Hypothyroidism](#) (an underactive thyroid gland)

Less common systemic causes include [leukemia](#) and [lymphoma](#) (blood cancers).

Disorders of the nervous system

Irritation of sensory nerves, for example, when a nerve is compressed, can cause itching localized to the part of the body supplied by that nerve. In contrast, some disorders that affect the nervous system may cause widespread (generalized) itching because itch neurons (one type of nerve cell) are too active or are less inhibited by other neurons. For example, in [multiple sclerosis](#), nerve conduction is impaired because of an autoimmune process where the immune system attacks the fatty tissue that coats nerve fibers (myelin). This process makes the itch neurons less inhibited and thus more active. Itch that results from nervous system disorders is called neuropathic itch.

Mental health conditions

Some people with mental health conditions may have itching for which no physical cause can be found. This type of itching is called psychogenic itching.

Drugs, medications, and chemicals

Drugs, including medications, and chemicals can cause itching when taken internally or when applied to the skin. Usually the itching is caused by an allergic reaction.

A few medications, such as morphine and some radiopaque contrast agents used when taking certain x-rays, can also cause itching without causing an allergic reaction.

Evaluation of Itching

Not every episode of itching requires immediate evaluation by a doctor. The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation. Most conditions that cause itching are not serious.

Warning signs

The following may indicate that the cause could be serious:

- Weight loss, fatigue, or night sweats may indicate a serious infection or a tumor.
- Weakness, numbness, or tingling may indicate a nervous system disorder.
- Abdominal pain or a yellowish discoloration of the skin and eyes (jaundice) may indicate a gallbladder or liver disorder.
- Excessive thirst, abnormally frequent urination, and weight loss may indicate diabetes.

When to see a doctor

People who have weight loss, fatigue, or night sweats should see a doctor as soon as possible.

People who have any of the other warning signs or severe itching should probably see a doctor immediately.

What the doctor does

Doctors ask many questions and look at the skin. Often, a person needs to undress so that the entire skin surface can be checked.

If no clear cause is found after checking the skin, doctors may do a complete physical examination to check for systemic causes. Testing may be necessary to diagnose certain systemic causes and sometimes skin disorders.

If itching is widespread and begins shortly after use of a drug or medication, that drug or medication may be the cause. If itching (usually with a rash) is confined to an area that was in contact with a substance, particularly if the substance is known to cause [contact dermatitis](#), that substance is a likely cause. However, allergic causes of widespread itching can be difficult to identify because affected people have usually eaten several different foods and have been exposed to many substances that could cause an allergic reaction before itching develops. Similarly, identifying a medication or drug that is causing the reaction in a person taking several medications or drugs may be difficult. Sometimes the person has been taking the drug or medication causing the reaction for months or even years before a reaction occurs.



Some Causes and Features of Itching

Cause	Common Features*	Tests
Skin causes		
Atopic dermatitis (sometimes called eczema)	Dryness, redness, and sometimes thickening and scaling, often in the folds of the elbows or behind the knees Usually a family history of allergies or rashes	A doctor's examination
Contact dermatitis	Redness and sometimes blisters in a shape or location corresponding to the substance causing the reaction (such as along the hairline when caused by hair dyes, on the wrist when caused by a watch, or on exposed skin when caused by poison ivy)	A doctor's examination
 Dry skin	Dry, itchy, scaly skin, usually on the legs, that is worsened by dry heat and develops or becomes worse in winter, after a hot bath, or after prolonged exposure to water	A doctor's examination
Fungal skin infections , such as ringworm , jock itch , or athlete's foot	A circular rash with raised scaly borders	Sometimes examination of skin



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